

SQL Queries and Answers (1–55)

20. Names & jobs separated by hyphen:

```
SELECT CONCAT(UCASE(LEFT(ename,1)), LCASE(SUBSTRING(ename,2)), ' - ',  
UCASE(LEFT(job,1)), LCASE(SUBSTRING(job,2))) AS emp_details FROM emp;
```

21. Empno, name, hiredate in dd/mm/yy for dept 20:

```
SELECT empno, ename, DATE_FORMAT(hiredate, '%d/%m/%y') FROM emp WHERE deptno = 20;
```

22. Months PRESIDENT worked:

```
SELECT TIMESTAMPDIFF(MONTH, hiredate, CURDATE()) FROM emp WHERE job='PRESIDENT';
```

23. Day SMITH joined:

```
SELECT ename, DAYNAME(hiredate) FROM emp WHERE ename='SMITH';
```

24. Time ADAMS joined:

```
SELECT ename, TIME(hiredate) FROM emp WHERE ename='ADAMS';
```

25. Day of month KING joined:

```
SELECT ename, DAY(hiredate) FROM emp WHERE ename='KING';
```

26. Month MARTIN joined:

```
SELECT ename, MONTHNAME(hiredate) FROM emp WHERE ename='MARTIN';
```

27. Quarter employees joined:

```
SELECT empno, ename, QUARTER(hiredate) FROM emp;
```

28. ANALYST formatted hiredate:

```
SELECT ename, DATE_FORMAT(hiredate,'The %D of %M %Y') FROM emp WHERE job='ANALYST';
```

29. Job classification:

```
SELECT ename, job,
```

```
CASE job WHEN 'CLERK' THEN 'A' WHEN 'MANAGER' THEN 'B'
WHEN 'PRESIDENT' THEN 'C' ELSE 'D' END AS job_class FROM emp;
```

30. Longest employee name length:

```
SELECT MAX(LENGTH(ename)) FROM emp;
```

31. Length of service:

```
SELECT ename, CONCAT(TIMESTAMPDIFF(YEAR,hiredate,CURDATE()),' years ',
MOD(TIMESTAMPDIFF(MONTH,hiredate,CURDATE()),12),' months') FROM emp;
```

32. Employees joined in 1985:

```
SELECT COUNT(*) FROM emp WHERE YEAR(hiredate)=1985;
```

33. Employees joined each month (1985):

```
SELECT MONTHNAME(hiredate), COUNT(*) FROM emp WHERE YEAR(hiredate)=1985
GROUP BY MONTH(hiredate) ORDER BY MONTH(hiredate);
```

34. Employees joined in March 1985:

```
SELECT COUNT(*) FROM emp WHERE YEAR(hiredate)=1985 AND
MONTHNAME(hiredate)='March';
```

35. Total sales:

```
SELECT SUM(amount) FROM sales;
```

36. Customer-wise low/high sales:

```
SELECT custid, MIN(amount), MAX(amount) FROM sales GROUP BY custid;
```

37. Product-wise low/high/total sales:

```
SELECT prodid, MIN(amount), MAX(amount), SUM(amount) FROM sales GROUP BY prodid;
```

38. Dept-wise avg salary (>3 employees):

SELECT deptno, AVG(sal) FROM emp GROUP BY deptno HAVING COUNT(*)>3;

39. Customer-wise sales except TKB SPORT SHOP with <=4 purchases:

SELECT custid, SUM(amount) FROM sales WHERE custid<>'TKB SPORT SHOP'
GROUP BY custid HAVING COUNT(*)<=4;

40. Salary stats:

SELECT MAX(sal), MIN(sal), SUM(sal), AVG(sal) FROM emp;

41. Salary stats per job:

SELECT job, MAX(sal), MIN(sal), SUM(sal), AVG(sal) FROM emp GROUP BY job;

42. Names with salary < avg salary dept 20:

SELECT ename FROM emp WHERE sal<(SELECT AVG(sal) FROM emp WHERE deptno=20);

43. Avg annual salary per job per dept:

SELECT deptno, job, AVG(sal*12) FROM emp GROUP BY deptno, job;

44. Salary/commission count dept 30:

SELECT COUNT(sal), COUNT(comm) FROM emp WHERE deptno=30;

45. Avg/min/max for CLERK & MANAGER:

SELECT job, AVG(sal), MIN(sal), MAX(sal) FROM emp
WHERE job IN('CLERK','MANAGER') GROUP BY job;

46. 2nd highest salary:

SELECT MAX(sal) FROM emp WHERE sal<(SELECT MAX(sal) FROM emp);

47. Employees earning max salary in their dept:

SELECT empno, ename, deptno, sal FROM emp e WHERE sal=(
SELECT MAX(sal) FROM emp WHERE deptno=e.deptno);

48. Salesman with highest total sales:

```
SELECT repid FROM sales GROUP BY repid ORDER BY SUM(amount) DESC LIMIT 1;
```

49. Highest salary per job:

```
SELECT job, MAX(sal) FROM emp GROUP BY job;
```

50. Most recently hired employee per dept:

```
SELECT empno, ename, deptno, hiredate FROM emp e WHERE hiredate=(  
SELECT MAX(hiredate) FROM emp WHERE deptno=e.deptno);
```

51. Year with most hires:

```
SELECT YEAR(hiredate), COUNT(*) FROM emp GROUP BY YEAR(hiredate)  
ORDER BY COUNT(*) DESC LIMIT 1;
```

52. Employee names occurring once:

```
SELECT ename FROM emp GROUP BY ename HAVING COUNT(*)=1;
```

53. Dept details + employee count:

```
SELECT d.*, COUNT(e.empno) FROM dept d  
LEFT JOIN emp e ON d.deptno=e.deptno GROUP BY d.deptno;
```

54. Departments with no employees:

```
SELECT d.deptno, d.dname FROM dept d LEFT JOIN emp e ON d.deptno=e.deptno  
GROUP BY d.deptno, d.dname HAVING COUNT(e.empno)=0;
```

55. Number of employees joining each month:

```
SELECT MONTHNAME(hiredate), COUNT(*) FROM emp GROUP BY MONTH(hiredate)  
ORDER BY MONTH(hiredate);
```